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TRENCHLESS REHABILITATION TECHNOLOGIES FOR UNDERGROUND CONDUITS

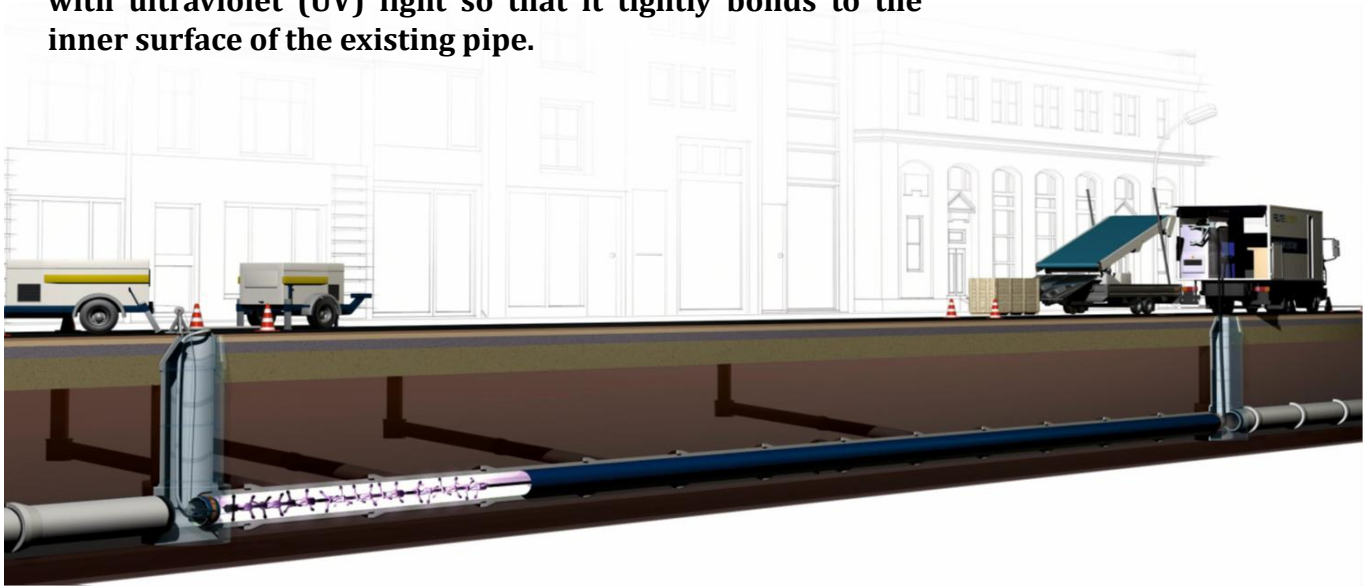
- CIPP Ultraviolet (UV) Technology
- PVC FOLD & FROM Technology
- SPRAYING Technology
  (Drinking Water- Pressure Pipes)
- HOSE LINER Technology
  (Drinking Water- Pressure Pipes)



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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

It is the process of installing a resin-impregnated glass fiber woven liner between two existing manholes and curing it with ultraviolet (UV) light so that it tightly bonds to the inner surface of the existing pipe.

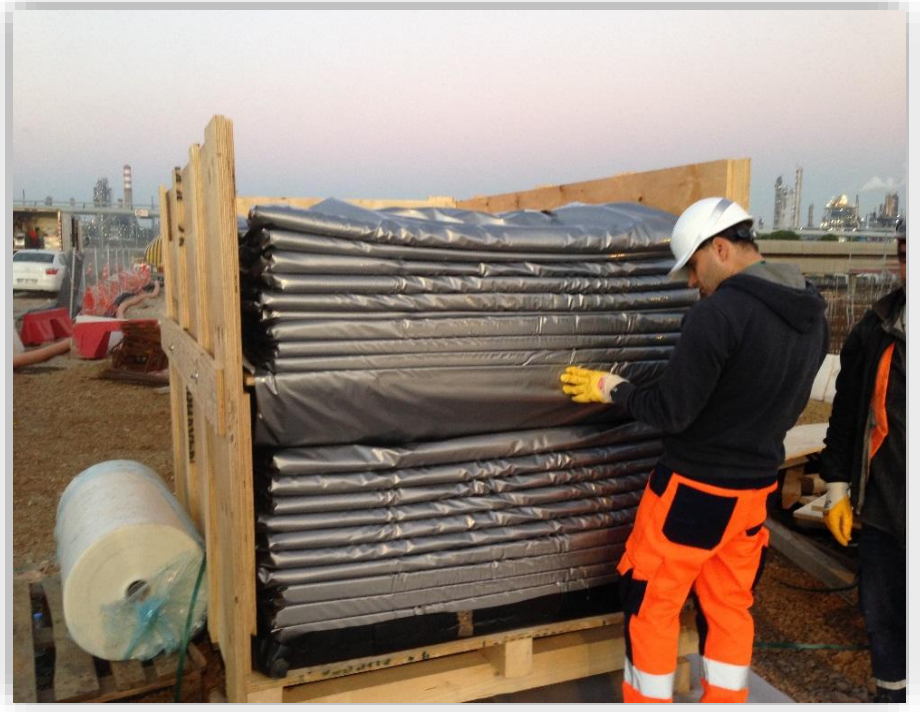


- Leading Organization in Trenchless Technologies
- Pipe Rehabilitation since 2004
- Experienced and trained staff
- State-of-the-art machinery and equipment
- 850,000 meters of pipe rehabilitation

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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

ADVANTAGES OF CIPP UV TECHNOLOGY



GLASS FIBER REINFORCED POLYESTER RESIN SYSTEM MATERIAL

- Today, in UV-light curing applications of CIPP materials, polyester or vinyl ester type resins are used together with glass fiber reinforcement. The combination of both forms a strong composite structure.
- High structural strength and stability (high ring stiffness; strong and durable resistance to pressure, impact, and tensile forces)
- Thin wall thickness (especially minimal loss of flow capacity in small-diameter pipelines)
- Fast installation and on-site application (rapid installation and commissioning in restricted areas, heavy traffic zones, or airport operations)
- Advanced and reliable quality control (resin impregnation is carried out in a factory environment, ensuring a high-quality and uniform liner)
- Environmentally friendly curing process (no waste generation during curing)
- Easy storage and handling (liners are transported to the site ready for installation directly from the production facility; unlike other CIPP systems, no separate resin impregnation process is required)

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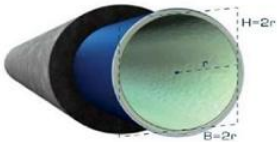
CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

GLASS FIBBER REINFORCED COMPOSITE LINING MATERIAL

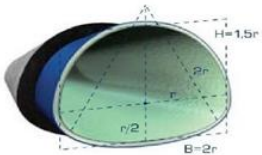
APPLICATION AREAS

- Wastewater and rainwater pipes
- Pressure pipes (drinking water, etc.)
- Gas pipes
- Industrial pipes (petrochemical, etc.)

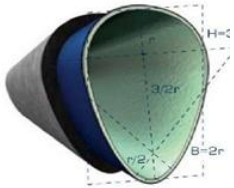
APPLICATION RANGE DN100 – DN2000



Circular shape



Horseshoe section



Egg shape

Mechanical properties

			Alphaliner	
	Norm	Unit	500G	1800H
Density	DIN EN ISO 1183-1	g/cm³	1,50	1,60
Glass fibre content (mass%)	DIN EN ISO 1172	(av.) %	46 ± 5	56 ± 5
E-modulus short-terme value (5% quantile)	DIN EN ISO 178	MPa	8.500	19.062
E-modulus short-terme value (5% quantile)	DIN EN ISO 1228 / ISO 7685	MPa	9.776	21.209
Flexural strength short-terme val. (5% qua.)	DIN EN ISO 178	MPa	180	320
Reduction ratio	DIN EN 761	-	1,60	1,31
Long-term bending E-modulus		MPa	5.312	14.551
E-modulus long-terme value (5% quantile)	DIN EN 1228	MPa	6.110	16.190
Long-term tensile bending stress		MPa	112	244
Material class	DWA-M 144-3	-	13	25

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BYPASS APPLICATION BEFORE LINING

BYPASS

The flow in the pipe line is interrupted by “Channel Blocking Pads”, and the flow is transferred to another line by excavation or with the help of pumps.







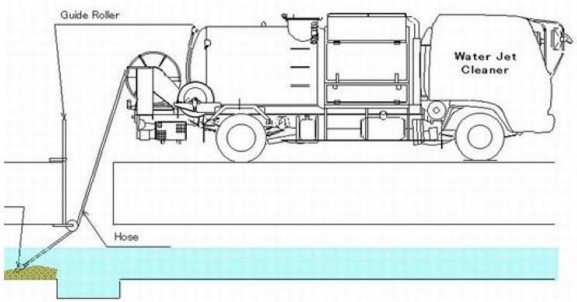
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CLEANING THE PIPELINE BEFORE LINING

CLEANING

High-pressure, jet duct combination cleaning with manual operation.
Before lining, thorough cleaning of the pipeline is crucial for a healthy coating. High-pressure jet duct combination systems are the most effective cleaning method.

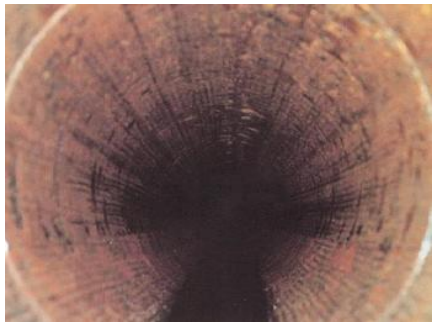




BEFORE



AFTER



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CCTV CAMERA RECORDING

CCTV

The pipelines are visualized using a CCTV channel inspection robot. An accurate diagnosis is made and reported to determine the appropriate repair method.





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INSTALLATION OF CIPP DN1000-DN2000 MATERIAL IN THE PIPELINE

Due to its heavy and bulky nature, UV Liner material is pulled into the existing pipeline using special machinery and equipment, and with the aid of cranes.







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PRESSURIZING THE MATERIAL

The UV Liner material, which is currently being applied to the pipeline, is inflated by being pressurized with the help of a special blower.













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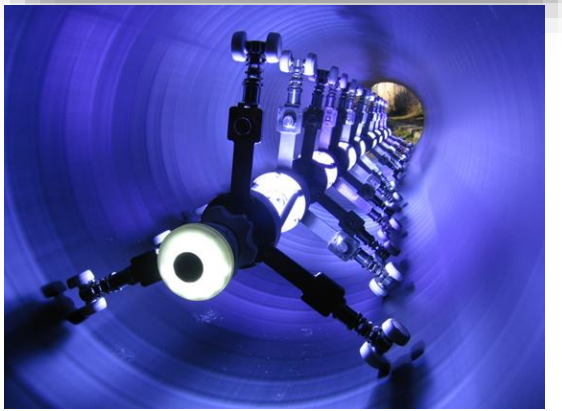
CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

CURING WITH LIGHT CHAIN

The inner surface of the pipe is cured using a lighttrain with Ultraviolet (UV) light, ensuring a homogeneous and tight lining that surrounds the pipe.







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CIPP – UV LINING STAGES - SUMMARY



BEFORE



PULLING OF THE LINER INTO THE HOSTPIPE

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CIPP – UV LINING STAGES - SUMMARY



CURING WITH LIGHT CHAIN

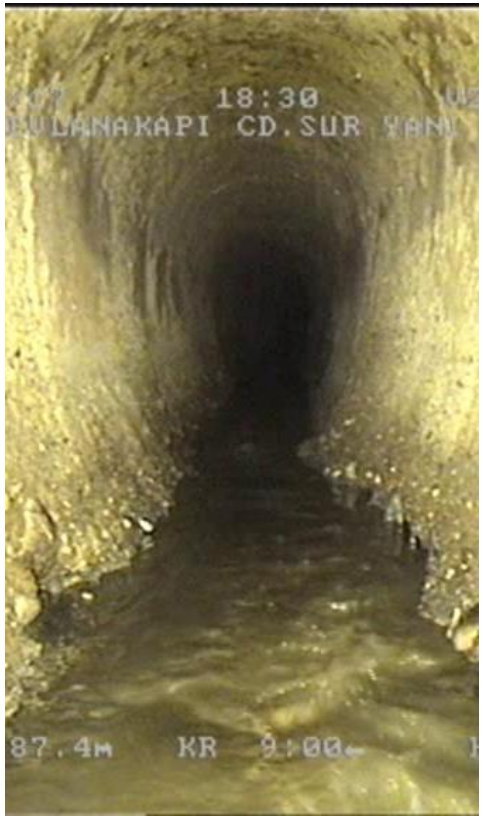


AFTER

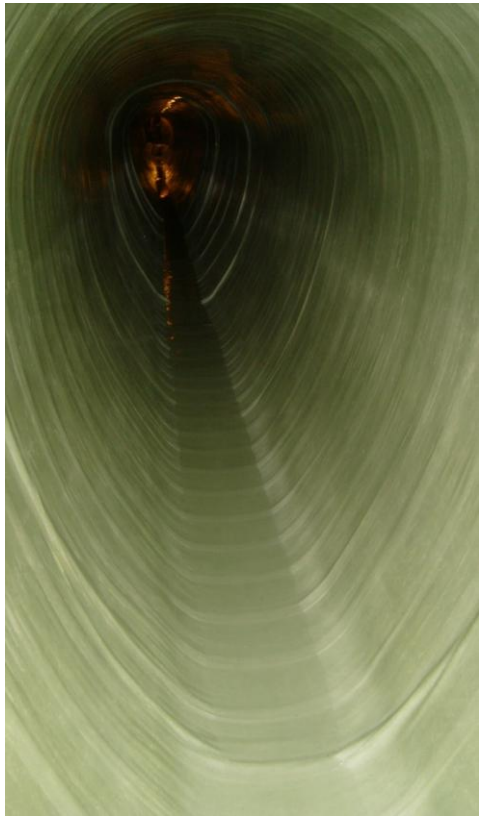
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EGG SHAPE





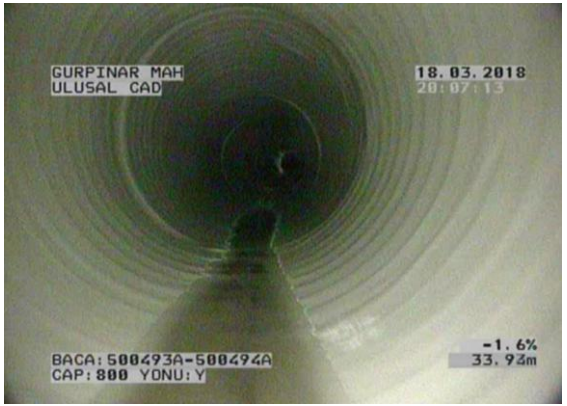


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BIG DIAMETERS (BIGGER THAN DN600)







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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

ELBİSTAN DN 1000 – CIPP UV

BEFORE



AFTER



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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

ELBİSTAN DN 1000 – CIPP UV

BEFORE



AFTER

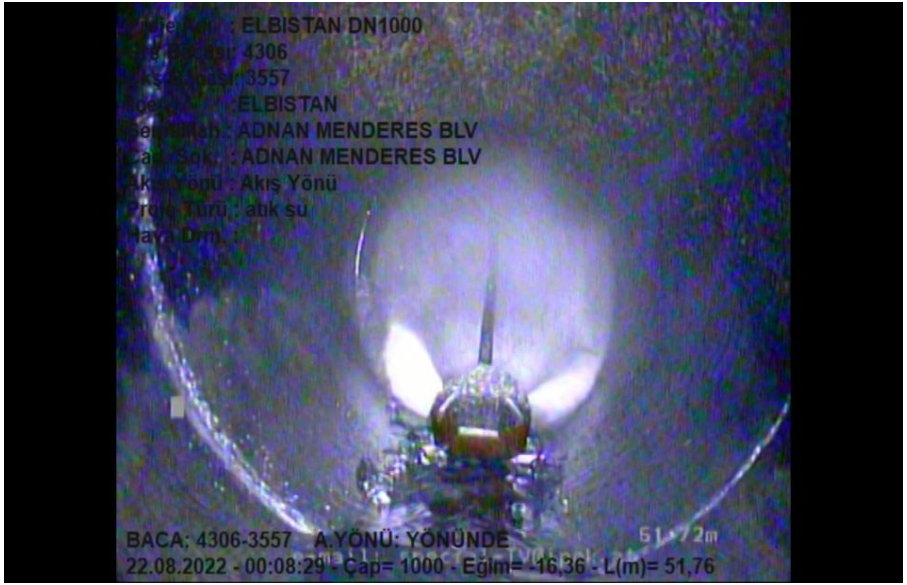


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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

IN WASTEWATER AND DRINKING WATER LINES – DN 1000

BEFORE



AFTER



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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

EGG SHAPE – DN 1000

BEFORE



AFTER



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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

DN 1400

BEFORE



AFTER



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CURED IN PLACE PIPE ULTRAVIOLET TECHNOLOGY

DN 1400

BEFORE



AFTER





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